

torch.nn.functional.one_hot#

https://pytorch.org/docs/stable/generated/torch.nn.functional.one_hot.html

Description:

Takes LongTensor with index values of shape (*) and returns a tensor of shape (*, num_classes) that have zeros everywhere except where the index of last dimension matches the corresponding value of the input tensor, in which case it will be 1. See also One-hot on Wikipedia . tensor (LongTensor) – class values of any shape. num_classes (int, optional) – Total number of classes. If set to -1, the number of classes will be inferred as one greater than the largest class value in the input tensor. Default: -1 LongTensor that has one more dimension with 1 values at the index of last dimension indicated by the input, and 0 everywhere else.

Function Signature

```
torch.nn.functional.one_hot(tensor, num_classes=-1) → LongTensor#
```

Parameters

Returns

Returns:

LongTensor that has one more dimension with 1 values at the index of last dimension indicated by the input, and 0 everywhere else.

Code Examples

```
>>> F.one_hot(torch.arange(0, 5) % 3)
tensor([[1, 0, 0],
        [0, 1, 0],
        [0, 0, 1],
        [1, 0, 0],
        [0, 1, 0]])

>>> F.one_hot(torch.arange(0, 5) % 3, num_classes=5)
tensor([[1, 0, 0, 0, 0],
        [0, 1, 0, 0, 0],
        [0, 0, 1, 0, 0],
        [1, 0, 0, 0, 0],
        [0, 1, 0, 0, 0]])

>>> F.one_hot(torch.arange(0, 6).view(3,2) % 3)
tensor([[[1, 0, 0],
         [0, 1, 0]],
        [[0, 0, 1],
         [1, 0, 0]],
        [[0, 1, 0],
         [0, 0, 1]]])
```

Detailed Documentation

Documentation

Takes LongTensor with index values of shape (*) and returns a tensor of shape (*, num_classes) that have zeros everywhere except where the index of last dimension matches the corresponding value of the input tensor, in which case it will be 1.

See also One-hot on Wikipedia .

tensor (LongTensor) – class values of any shape.

num_classes (int, optional) – Total number of classes. If set to -1, the number of classes will be inferred as one greater than the largest class value in the input tensor. Default: -1

LongTensor that has one more dimension with 1 values at the index of last dimension indicated by the input, and 0 everywhere else.